

### 3 Divide and Conquer

Search for 64.

```
10 20 37 43 52 76 85 93
10 20 37 *43* | 52 76 85 93 [43 vs 64]
52 *76* | 85 93 [76 vs 64]
*52* | 76 [52 vs 64]
```

Three comparisons.

#### 3.1 Merge Sort

```
38 27 43 3 | 9 82 10
38 27 | 43 3      9 82 | 10
38 | 27      43 | 3      9 | 82      10
38      27      43      3      9      82      10
27 38      3 43      9 82      10
3 27 28 43      9 10 82
3 9 10 38 43 82
```

#### 3.2 Substitution Method

$2^n - 1$

**Base case:**  $2 - 1 = 1$

**Inductive:**

(1)

$$T(n) = 2^n - 1 \quad (2)$$

$$T(n+1) = 2^{n+1} - 1 \quad (3)$$

$$= 2 T(n) + 1 \quad (4)$$

$$= 2(2^n - 1) \quad (5)$$

$$= 2^{n+1} - 1 \quad (6)$$